Governance operators and viability kernels of the Edwards Aquifer: a J-17 test

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Abstract

Problem. Permitted groundwater systems manage drought with triggers, but competing rules are rarely compared against a common criterion for robust protection and permitted supply.

Approach. We compute robust viability kernels for a declared family of pumping rules on the 1934–2023 J-17 annual-mean head record, using training-mean pumping (282.16×10^3 acre-ft yr^{-1}), flat caps, an annual-mean Stage-I sketch, and a critical-period-management cascade under specified persistent recharge floors. Retention follows a dated protocol frozen before scoring, comparing protection with training-mean pumping and supply with the smallest securing flat cap.

Results. Under the perpetual drought-of-record floor at the 618-ft physical threshold, the training-mean policy's worst-case attractor is 615.72 ft and its kernel empties beyond 13 years; flat cuts of 10% or deeper and both reactive rules make the declared safe set invariant, with a smallest securing cut interpolated at 7.2% of training-mean pumping (31.5% of current use). The reactive rules meet the retention rule only nominally at 618 ft under the mildest floor class: their +3.3% (Stage-I sketch) and +0.4% (cascade) supply margins rely on the hybrid replay score and are not robust certificates. Positive-pumping certified kernels are empty beyond three years under the declared defect erosion. At 660 ft, the frozen retention rule retains no policy under the persistent drought-of-record floor. For the point-fit coefficients, the safe set is not invariant even under zero pumping; a post-freeze 5,000-refit residual bootstrap places the zero-pumping attractor at or above 660 ft in 344 refits (6.88%; 5th–95th percentiles 628.99–661.16 ft under UC-min), so the point-fit positive-margin premise is not robust to coefficient uncertainty.

Implications. The 660-ft result is conditional on the fitted one-pool map, annual-mean policy sketch, and persistent-floor class. The post-freeze bootstrap is a coefficient-sensitivity check conditional on the training window, iid residual resampling, and fixed UC-min floor — not a probability of field safety—and does not re-score the frozen retention rule. Neither result estimates the implemented 10-day Critical Period Management rule's effect on stage frequency or springflow. The transferable contribution is the scoring protocol; the numerical verdicts are limited to this J-17 map, policy family, and disturbance specification.

Keywords: Edwards Aquifer; critical period management; J-17 index well; pumping policies; robust viability; viability kernel; groundwater governance

Article Impact Statement. This study demonstrates a frozen intervention-scoring protocol — robust viability kernels under declared recharge floors, defect-limited erosion, and an explicit retention rule — on the Edwards Aquifer's J-17 annual-mean record. At the 618-ft physical threshold, the reactive rules' +3.3% and +0.4% supply margins over the kernel-matched

flat-90% cap are nominal hybrid-score results, not robust supply guarantees. At 660 ft, the point-fit map admits no invariant safe set under the persistent drought-of-record floor, including at zero pumping. In the post-freeze bootstrap, the zero-pumping attractor is at or above 660 ft in 6.88% of refits, so the point-fit margin is not coefficient-robust. This model-conditional sensitivity is not a performance evaluation of the live 10-day CPM rule or of springflow outcomes. The method can be applied elsewhere after the model, disturbance class, policy family, and defect are re-specified; these J-17 verdicts do not transfer to other aquifer pools.

1. Introduction

Drought management in permitted groundwater systems is built on triggers. An index-well or springflow reading crosses a threshold, and pumping reductions activate. Trigger design has a long evaluation literature: drought indicators and triggers are properly assessed as stochastic objects, with explicit attention to how indicator levels, trigger thresholds, and drought stages interact (Steinemann 2003). Robustness has been a recurring concern of water-resources planning since the recognition that optimal solutions can be brittle to the hydrologic assumptions that produce them (Watkins and McKinney 1997). The operational question for a district that already operates triggers is narrower, and comparative: given a declared family of pumping rules, which rules deliver protection under adverse recharge, and at what cost in permitted supply?

The Edwards (Balcones Fault Zone) Aquifer of south-central Texas is the most fully institutionalized example of trigger-based groundwater governance in North America. Its critical-period management program stages pumping reductions at index-well thresholds. The reductions protect springflow at Comal and San Marcos Springs, whose endangered-species requirements are administered through the Edwards Aquifer Habitat Conservation Plan, the subject of a standing scientific review by the National Academies (National Research Council 2015). The aquifer's regional behavior has long been represented — and debated — through lumped and equivalent-porous-media models, from the Barton Springs groundwater availability model (Scanlon et al. 2003) to its recalibration by the Texas Water Development Board (Hutchison and Hill 2011). This study scores, rather than asserts, the protection supplied by a declared family of pumping rules on the measured J-17 record.

The instrument is the robust viability kernel: the set of initial heads from which a closed-loop pumping rule keeps the system inside a declared safe set under every disturbance in a declared class. Viability methods have an established lineage in natural-resource management, where constraints rather than objectives define the policy question (Krawczyk and Pharo 2013), and their computational tractability on management-scale problems is documented by fishery case studies (Krawczyk et al. 2013). The declared safe set here is a floor on the productive store rather than on a year's yield. The head threshold marks the level below which springflow at Comal and San Marcos approaches cessation. The policy question is therefore the protection of the base that regenerates the yield, and a trigger's function is to keep the store from being drawn down to that floor at the cost of near-term permitted supply. The governed object is the one-pool affine head map fitted in the published companion forecast-evaluation study on the same series (Abaee, 2026a). Its 7.55-ft one-step result conditions on realized annual fluxes and is a conditional diagnostic, not a forecast; the genuine one-year forecast scores were 13.23 ft for persistence and 12.84 ft for AR(1). This study asks the governance question that the forecast study could not: whether a declared pumping rule changes the viability kernel of the real system as represented by the fitted map, and at what cost in permitted supply. A companion intervention study on a marine fishery stock applies the same design and reaches the mirror verdict — none retained there, reactive rules retained here (nominally, under the mildest floor class); the mechanism contrast (fast wet-year recovery in this aquifer against a fatal productivity floor at the cod reference point) is the only reason to mention it, and the two systems' scores are never pooled (Abaee, 2026b).

The complete comparison — object, defect declaration, disturbance classes, policy family, and retention rule — was frozen in a dated protocol (2026-08-26) before any score was computed.

Related work

The regulatory object. The J-17 index well is not merely a monitoring point: it is the official regulatory trigger of the Edwards Aquifer Authority's Critical Period Management plan, and crossing its thresholds automatically imposes pumping restrictions (Edwards Aquifer Authority, 2024). Under the EAA Act, permitted withdrawals from the aquifer are capped, and the critical-period stages impose graduated withdrawal reductions on the San Antonio Pool as the J-17 level and Comal and San Marcos springflows decline; the Stage I trigger for the San Antonio Pool is a J-17 level below 660 ft MSL with a 20% withdrawal reduction, rising to a 40% reduction at Stage IV; the Uvalde Pool is staged separately on J-27 (Edwards Aquifer Recovery Implementation Program, 2021). The 660-ft level studied here matches the actual Stage I threshold, but the simulated policy is an annual-mean 20%-at-660 sketch. It does not reproduce the operational 10-day trigger algorithm or estimate its effect on springflow. The 660-ft comparison is therefore institutionally grounded but remains a model-based counterfactual, not a performance evaluation of the rule in force. The 618-ft level is the physical threshold at which the analysis finds protection to be a different question.

Viability applied to groundwater. The viability approach has been applied to aquifer management repeatedly, and the comparison it invites is by now standard: in the optimal-control framing an environmental objective enters as a monetised externality, while in the viable framing it enters as a constraint to be satisfied, and the two produce different trajectories and different long-run levels (Pereau, Pryet and Rambonilaza, 2019). The viability kernel has been identified analytically in hydro-economic models of aquifers under conflicting food-security and ecosystem objectives (Pereau, Mouysset and Doyen, 2018), and the wider survey literature records reservoir and groundwater management as established application areas (Oubraham and Zaccour, 2018).

What this paper adds is an applied comparison of a declared policy family under persistent recharge floors, with a dated retention rule and separate nominal and defect-eroded certificates. At the point-fit 660-ft threshold, no evaluated policy is retained: the reactive rules are invisible to a kernel defined at their own trigger, while flat caps extend finite-horizon viability without making the safe set invariant. This is a claim about the interaction of the declared trigger, disturbance class, and scoring rule — not a new theorem in viability theory, nor a general claim that prior literature has ignored trigger geometry. The post-freeze bootstrap in Section 3.7 further shows that the positive zero-pumping margin at 660 ft under UC-min is not robust to coefficient resampling.

From safe yield to sustainability. The classical safe-yield concept — balancing annual withdrawal against recharge — was shown to be unsustainable in principle because it ignores natural discharge already supported by recharge (Sophocleous, 1997; Alley, Reilly and Franke, 1999; Sophocleous, 2000; Alley and Leake, 2004); the persistence of that water-budget reasoning is also documented (Devlin and Sophocleous, 2005). This paper does not use "safe yield" as a model or policy category. Its narrower connection is mathematical: for a fixed contractive map with a negative pumping coefficient, a positive zero-pumping attractor-to- threshold margin rules out invariance under every nonnegative pumping sequence. At the point fit, that condition holds at 660 ft. The post-freeze UC-min bootstrap shows that its sign is not robust to coefficient resampling (Section 3.7). This conditional result is not evidence of physical inevitability and is not a test of the safe-yield doctrine; it is a map-specific comparison under an adverse recharge floor.

Viability theory. That the governing question is which states admit controls keeping every constraint satisfied over time, rather than which policy optimises, is the founding move of viability theory (Aubin, 1991); the obstruction calculus underlying the certificates used here is the observation-constrained development of that programme.

2. Methods

2.1 Object and dynamics

The object is that of the companion forecast evaluation: J-17 annual-mean head H_t (ft above mean sea level), San Antonio Pool, 1934–2023, drawn from the registered twenty-column analysis panel (that evaluation's fixed analysis dataset — the measured head, the constructed recharge and pumpage series, and the service and derived series declared there — of which only the head, recharge, and pumpage columns enter this analysis). The head is a measured well level, not an assessment inversion. Recharge R (USGS-estimated, Puente 1978 method; Umphres and Choi 2025) and pumpage P (Edwards Aquifer Authority Table 1) are the fluxes. No new data enter the analysis.

The dynamics are the causal stock-flow class of the companion forecast study, with one pool and affine form:

$$\Delta H_t = \alpha + \beta R_t + \gamma P_t + \delta H_{t-1}.$$

Definition 2.1 (One-pool affine head map). Let H_t denote the annual-mean head (ft above msl) at J-17, R_t the annual recharge, and P_t the annual pumpage. Under the hypotheses (F1) one-pool storage, (F2) affine dependence on (R_t, P_t, H_{t−1}), and (F3) ordinary-least-squares fitting on 1934–1990 (56 transitions), the head map is the relation above.

The fit is $\alpha = 163.49$, $\beta = 0.0198$ ft per 10^3 acre-ft, $\gamma = -0.02844$ ft per 10^3 acre-ft, $\delta = -0.2539$, so $a = 1 + \delta = 0.7461$. The contraction $a = 0.7461$ corresponds to 25.4% per year mean reversion (the companion's rolling AR(1) on the same series gives 0.66 — different window, different object; the full-sample coefficients there are (0.017, −0.026) against this window's (0.0198, −0.02844)). Residual SD is 5.60 ft; training maximum is 15.41 ft. Out of sample (1991–2023, audit only, no refitting), the SD is 8.40 ft and the maximum is 21.81 ft. The out-of-sample maximum exceeds the declared defect. This fact is recorded below and not repaired, per protocol. The coefficients are fitted on the largely unregulated 1934–1990 era, so ordinary least squares cannot separate the aquifer's physical response from the pumpers' historical drought reactions, and imposing state-dependent rules under those coefficients assumes the physics–behavior separation the estimator does not guarantee (Lucas 1976); this bears on whether γ may be read as a policy lever at all, and the companion found the sign of γ reversed on the 1934–1950 subwindow.

The information pattern is the following: the manager ends year t knowing (H_t, R_t, P_t) and sets P_{t+1} = π(H_t). The recharge R_{t+1} is unknown at decision time and is treated adversarially within a declared persistent floor. The kernels therefore assume year-end J-17 is known exactly; the actual CPM triggers on 10-day means and cuts within the year, so annual triggering overstates the lag and understates the cut-active fraction.

2.2 Uncertainty classes and safe sets

The disturbance classes are persistent recharge floors computed on the training window: UC-min = 43.7 (the 1956 drought-of-record year, held perpetually), UC-q05 = 166.5, and UC-q10 = 179.1 × 10^3 acre-ft yr^{-1}. UC-q05 and UC-q10 are the 3rd and 6th lowest recorded annual recharges of the 57 training years (linear-interpolation percentile estimator); the floors themselves have historical precedent, and only their perpetuity is harsher than any recorded year. They are certification geometry, not recharge forecasts.

The safe sets are the two declared normative thresholds, fixed in the frozen protocol: K^*_{phys} = 618 ft (Comal Springs cessation proximity) and K^*_{inst} = 660 ft (the post-2007 Stage I trigger, not applied to pre-2007 history; a 10-day mean in the actual institution, an annual mean here). The model domain is [610, 710] ft. Upward exits above 710 ft are model-domain exits, not threshold violations.

2.3 Governance family

The declared governance family is the following. Training-mean pumping (called BAU in the frozen protocol; the label is retained in tables and, under this declaration, in running prose as the family's baseline label) holds pumping at the 1934–1990 training mean, P $\equiv \bar{P}$ = 282.16 $\times$ 10^3 acre-ft yr^{-1} — a flat 100% cap on the unregulated mean, not the historical pumpage path and not current use. Flat caps prescribe $\rho\bar{P}$ with $\rho \in \{0.9, 0.8, 0.7, 0.6, 0.5, 0.0\}$. The Stage-I reactive rule (S1) cuts pumping 20% when H $<$ 660 ft; the reduction is a 20%-at-660 sketch of the Authority's published Stage I (Edwards Aquifer Authority, Critical Period / Drought Management; the actual rule keys on 10-day J-17 or Comal flow). The critical-period-management (CPM) cascade prescribes cumulative stage totals of 20/30/35/40% cuts at H $<$ 660/650/640/630 ft (cumulative, not stacked). The actual program declares five stages (Stage V at 625 ft); only the four-stage sketch is scored, Stage I is verified against the published 20%/660 pair, and stages II–IV are declared scenarios, not verified institutions.

The CPM supply figure below is the stage-weighted evaluation on the observed occupancies of the training era (1934–1990, n = 57 years): Stage I (H $<$ 660 ft) 20 years (35.1%), Stage II (H $<$ 650 ft) 12 (21.1%), Stage III (H $<$ 640 ft) 5 (8.8%), and Stage IV (H $<$ 630 ft) 1 (1.8%; 1956). The cumulative 20/30/35/40% cascade evaluated on these occupancies reproduces the replay's mean prescribed pumping exactly (254.93 $\times$ 10^3 acre-ft yr^{-1}). Out of sample (1991–2023, n = 33 years) the occupancies are 33.3%, 15.2%, 6.1%, and 0.0%. Stages II and III recur out of sample (15.2%, 6.1%); Stage IV does not (0.0%; 1956 alone is below 630 ft), so CPM's deepest cut is a 1956-only scenario in the historical measure.

Definition 2.2 (Robust viability kernel). Fix a policy π, a disturbance class $\mathcal{D}$, a safe set K^*, and a horizon T. The robust T-step viability kernel Viab_T(π, K^*, $\mathcal{D}$) is the set of initial heads from which the closed loop P_{t+1} = π(H_t) keeps H inside K^* under every recharge sequence in $\mathcal{D}$.

For each policy and disturbance class, the robust T-step viability kernel of the safe set is computed by iterating the worst-case closed loop. The nominal kernel is reported without erosion. The certified layer applies the defect-to-margin conversion in the form the fitted map admits. The term "kernel" is used in its closed-loop reading — the robust positively invariant set of a fixed declared policy — not in the classical viability reading of Aubin (1991), which is existential over controls. No control choice enters the kernel computation. Only the evaluation of declared rules under the disturbance floor enters. With the declared policies all deterministic functions of the observed head, each robust kernel reduces to the invariance statement of a one-dimensional closed loop — attractor and domain-top arithmetic rather than control synthesis. The word is retained because the object answers the same question ("from which states can the constraint be held?") in the policy-fixed form.

2.4 Retention rule and certified layer

The retention rule decides which non-training-mean policies earn their additional structure. The certified layer converts the declared defect into an erosion margin on the kernel.

Definition 2.3 (Retention rule). A non-BAU policy is retained only if (R1) its robust kernel is at least as protective as training-mean pumping's at every reading, compared on the kernel lower boundary with empty = worst, and (R2) it permits more pumping than the smallest securing flat cap with matched protection — on the declared 10% grid this resolves

as the largest-supply member of the matched class (flat-90%); the grid-resolution sensitivity is reported in Section 3.4 (Table 4).

The scoring regimes are declared here rather than left implicit. **Protection is scored under the disturbance classes** — the perpetual floors of Section 2.2. **Supply is scored as the 1934–1990 historical replay mean.** The retention verdict is therefore a hybrid: worst-case protection at historical-mean entitlement, not a within-scenario dominance. Under the floor classes alone, the reactive rules are identically their matched flat caps (the cut is active every year) and carry no supply margin. The retention decision is made at the nominal level and re-checked at the certified level. The certified re-check holds only over the horizons where the certified kernels are nonempty — a horizon-truncated object, stated as such whenever it is invoked. At 618 ft under UC-min, R1 is close to automatic: every cut of 10% or deeper in the family makes the safe set invariant, so retention reduces to R2 (supply at matched protection).

Calculation 2.1 (Certified erosion margin). Let $\varepsilon = 15.41$ ft denote the declared uniform defect (the training maximum residual) and let a = 0.7461 denote the closed-loop contraction rate. Under the hypothesis (F4) that the autonomous contraction rate applies uniformly to the residual, the T-step erosion margin is

$$r_T = \varepsilon \frac{1 - a^T}{1 - a},$$

giving $r_1 = 15.41$, $r_3 = 35.49$, $r_4 = 41.89$, $r_5 = 46.66$, $r_\infty = 60.70$ ft.

Derivation. Under (F4) the closed-loop residual obeys e_T = a^T e_0. Summing the geometric series of perturbations of magnitude at most ε yields r_T = $\varepsilon(1 + a + \ldots + a^{T-1}) = \varepsilon(1 - a^T)/(1 - a)$; the result is the triangle-inequality bound for bounded additive perturbations of a contracting scalar map. Numerical evaluation at the listed horizons gives the listed values. □

Definition 2.4 (Certified kernel). The certified kernel at horizon T is the nominal kernel of $K^* + r_T$, with r_T as in Calculation 2.1.

Two properties of the certified layer must be stated before the results, because the audit correctly flagged the earlier account as imprecise. First, in this affine family every policy enters the map only through the intercept (γP is a constant shift), so the contraction rate a is identical across policies and (F4) is exact, not an approximation: the certified kernel is the closed-loop nominal kernel of a raised threshold, feedback included. Second, the mechanism by which the reactive rules collapse at the certified level is trigger-band entry: the eroded threshold $K^* + r_T$ crosses the 660-ft trigger between T = 4 (618 + 41.89 = 659.89) and T = 5 (618 + 46.66 = 664.66), so any threshold-erosion certificate makes every below-660 trigger rule identical to training-mean pumping from T ≥ 4 — the same trigger-on-boundary invisibility as Section 3.2, re-entering by construction. The constant-r_T form is also conservative by construction: a per-step tube $K^* + r_t$ (t = 1..T) is tighter than $K^* + r_T$ held at every step.

3. Results

3.1 Worst-case attractors and the minimal cut

Table 1. Worst-case attractors of the closed loop (ft), by policy and recharge floor.

Policy	UC-min	UC-q05	UC-q10
training-mean pumping (BAU)	615.72	625.31	626.29
flat-90%	618.88	628.47	629.45
flat-80%	622.04	631.63	632.61
S1 (20% cut < 660 ft)	622.04	631.63	632.61
flat-70%	625.20	634.79	635.77
CPM cascade	628.36	636.37	637.35

Policy	UC-min	UC-q05	UC-q10
flat-60%	628.36	637.95	638.93
flat-50%	631.52	641.11	642.09
flat-0 (zero pumping)	647.32	656.91	657.90

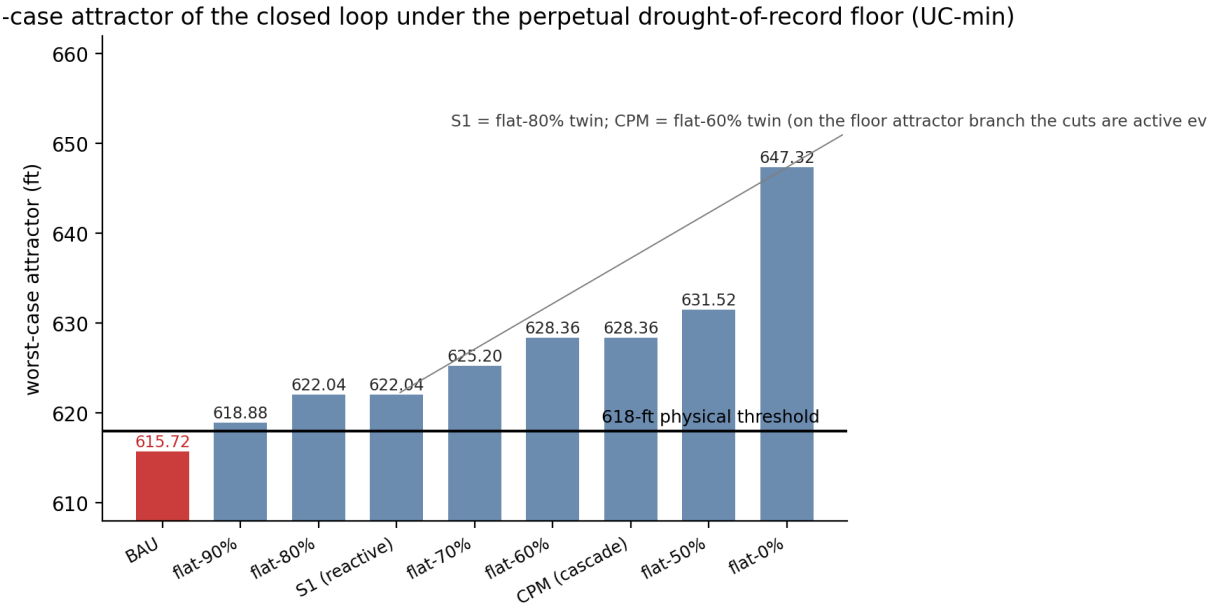


Figure 1: Worst-case attractor of the closed loop under the perpetual drought-of-record floor (UC-min) by policy, from the registered kernel computation (Table 1). Training-mean pumping's attractor (615.72 ft) sits below the 618-ft physical threshold. Every cut policy — flat caps of 10% and deeper, the Stage I reactive rule, and the critical-period-management cascade — holds its attractor at or above the threshold. Under UC-min the reactive rules coincide with their attractor twins (S1 = flat-80%, CPM = flat-60%; under UC-q05 the CPM attractor sits between the Stage III and IV triggers and equals a ~65% cap); flat-90% clears the threshold by 0.88 ft — a margin the bootstrap of Section 3.7 shows is not resolved by the fit.

Under a perpetual 1956-recharge floor, training-mean pumping's attractor (615.72 ft) sits below the physical threshold. The smallest flat cut whose attractor clears 618 ft is an interpolated **7.2%** of training-mean pumping, which lies outside the declared family (on the current-mean baseline of 382.16×10^3 acre-ft yr^{-1} the same arithmetic gives a securing cut of about 31.5%, and the Stage-I 20% cut does not secure — Section 3.4). The Stage-I reactive rule's attractor equals flat-80%'s, because the cut is active on the entire attractor branch.

3.2 Nominal kernels

Table 2. Nominal kernel lower boundary (ft) at the 618-ft threshold under UC-min, by horizon. Cut policies (flat-90% and deeper, S1, CPM) hold the whole safe set [618, 710] at every horizon, including the infinite horizon.

Horizon T	1	5	10	12	12.7 (crossover)	∞
BAU boundary	618.8	625.6	658.4	692.6	710	empty

At the physical threshold (618 ft) under UC-min, training-mean pumping's kernel boundary climbs 618.8 (T = 1) → 625.6 (T = 5) → 658.4 (T = 10), and the kernel is empty beyond T ≈ 13 years. The emptiness is a domain-top event, and its arithmetic should be read exactly: the

boundary $B(T) = H^* + (618 - H^*)/a^T$ reaches the declared safe-set ceiling when the *required initial head* exceeds 710 ft — trajectories fall toward the attractor, they do not exit upward. The horizon is a function of the arbitrary ceiling: $T_{\text{empty}}(C) = \ln\big((C-H^*)/(K^*-H^*)\big)/\ln(1/a)$, so the 710-ft ceiling gives 12.7 years, the observed maximum annual mean (691.96 ft, the 1992 mean) gives 12.0 years, and with no ceiling the kernel is never empty in the domain sense — every state converges to the attractor, and "empty beyond T" means "no initial head in the domain lasts T years." With the ceiling stated, business-as-usual is not robustly viable against a perpetual drought-of-record, and the observed 1992 mean (691.96 ft) sits almost at the T = 12 boundary. Every cut policy in the family — a 10% flat cap, S1, CPM, and deeper — makes [618, 710] robustly invariant: the whole declared safe set is the kernel at every horizon, including the infinite horizon. Under UC-q05/UC-q10 the safe set is already invariant at training-mean pumping (attractors 625.3/626.3 ft); governance differentiates the kernel only under the drought-floor class.

At the institutional threshold (660 ft), the point-fit result is a **design observation about scoring a trigger against its own level**, and the kernel statement splits by policy class. For the **reactive rules** (S1, CPM) the equality with training-mean pumping is exact at every horizon: the boundaries (675.1 at T = 1, 695.3 at T = 2 under UC-min; empty from T = 3) lie strictly above the first-stage trigger (660 ft, the highest of the four declared trigger levels), no reactive rule fires in the viable region, and the kernel is policy-invariant for them. That equality is a geometric identity — a trigger placed on the boundary of a constraint is invisible to the viability kernel of that same constraint, and no rule that cuts only below 660 ft can improve the invariance of {H ≥ 660} — and it is a category observation about the scoring, not a deficiency of the rules: the cascade's institutional purpose is springflow protection near 618 ft, not head invariance at 660 ft. For the **flat caps** the equality is false. They move the finite-horizon boundaries (flat-80% reads ≈673 at T = 1 against BAU's 675.1) and extend the viable horizon (from T = 3 at BAU through T ≈ 6 at flat-50% to T ≈ 6–11 for zero pumping by the same ceiling arithmetic; zero pumping's attractor 647.3/657.9 ft still sits below 660 ft). Demand management therefore extends the viable **horizon, not invariance**, and the correct point-fit statement of the 660-ft reading is: no policy in the declared family, zero pumping included, holds {H ≥ 660} invariant under the drought floors — the threshold is protected by wet years, not by the pumping family. The coefficient sensitivity of this point-fit conclusion under UC-min is assessed separately in Section 3.7. This is the frequency-management rationale the actual CPM rule implements, and that rationale is outside the robust-kernel frame.

The negative reading extends beyond "the declared family"; stating the stronger form makes its condition and scope explicit. Write the fitted map as $H_{t+1} = aH_t + \alpha + \beta R_t + \gamma P_t$ with $0 < a < 1$ and $\gamma < 0$ (here $a = 1 + \delta = 0.7461$ and $\gamma = -0.02844$, from the Section 2.2 fit; both are fitted signs, reported rather than assumed). Because $\gamma < 0$, any non-negative pumping sequence $P_t \geq 0$ gives $H_{t+1} \leq aH_t + \alpha + \beta F$ pointwise under a recharge floor F; because $0 < a < 1$, that comparison map contracts to the zero-pumping attractor $H_0^* = (\alpha+\beta F)/(1-a)$. So whenever the *attractor-to-threshold margin* $m = K - H_0^*$ is positive, every non-negative pumping rule — not merely every member of the declared family — fails to hold $\{H \geq K\}$ invariant. At the declared point coefficients, the margin is positive under all three floors, which is why zero pumping fails there. Whether this positive-margin condition survives coefficient uncertainty at UC-min is examined in Section 3.7.

Two consequences follow. First, the argument quantifies over pumping *sequences*, so it is indifferent to a rule's functional form: it covers rules that respond to several state variables at once, and rules with hysteresis (a stage entered on one indicator and exited only on another), neither of which the declared family here can express. A district operating a richer rule set inherits this negative verdict only conditional on the point-fit coefficients and positive-margin hypothesis; the UC-min coefficient sensitivity in Section 3.7 limits that conclusion. Second, the converse does not hold. A negative margin says only that zero pumping *could* hold K; whether

some member of a particular declared family does so is a separate, family-relative question, and one that a larger family can answer differently. Emptiness generalizes; construction does not.

3.3 Certified kernels

With the erosion of Section 2.4 applied, every positive-pumping policy in the family has an empty certified kernel beyond **T = 3 years** at the physical threshold and beyond T = 1 year at the institutional threshold. Zero pumping is the exception. Its certified physical-threshold kernel is nonempty through T = 4 under UC-min (lower boundary ≈687.9 ft from the certified-kernel algebra; the registered horizon grid {1, 2, 3, 5, 8, 10, 15, 20, ∞} does not sample T = 4 — the algebra is the record there and the grid is annotated accordingly) and through T = 5 under UC-q05/UC-q10. The certified boundaries at T = 3, UC-min, 618 ft: flat-0 662.2 < flat-80 697.8 < BAU = S1 = CPM 706.7 ft. The certified-level inference is drawn explicitly: at the one certified horizon tabulated, the reactive rules inherit training-mean pumping's boundary (706.7) — strictly above flat-80%'s (697.8) — so certified dominance of S1 over its matched flat cap fails. The +3.3%/+16.2% supply margins are nominal-level comparisons. Certified retention is a different, horizon-truncated object from nominal retention (Section 2.4).

The bound is optimistic, and the audit's sensitivity is recorded: at the out-of-sample defect (ε = 21.81 ft) the erosion margins scale to r_3 = 50.22, r_5 = 66.04, r_∞ = 85.90 ft, so certified emptiness beyond T = 3 is a *lower* bound on how fast certification dies, and certified horizons do not appear as findings beyond this section's scope statement. The binding constraint on certified intervention claims is the model defect, not the governance. The certified comparison is also conservative in form against the state-dependent rules for the geometric reason of Section 2.4 — the eroded threshold enters the trigger band from T = 4 — which is a property of threshold-erosion certificates generally, not of these rules.

3.4 Supply and retention

Table 3. Mean prescribed pumping: the frozen open-loop replay (actual observed heads, 1934–1990) and the post-freeze closed-loop replay (head simulated by the map under actual recharge, same policy in force), 10^3 acre-ft yr^{-1}.

Policy	Open-loop supply (frozen)	Cut active	Closed-loop supply (post-freeze)	Closed-loop cut active
BAU (training-mean)	282.16	0%	282.16	0%
flat-90%	253.94	100%	253.94	100%
flat-80%	225.73	100%	225.73	100%
flat-70%	197.51	100%	197.51	100%
S1	**262.36**	35.1%	258.98	41.1%
CPM	**254.93**	35.1%	254.70	41.1% (any stage)
flat-60%	169.29	100%	169.30	100%
flat-50%	141.08	100%	141.08	100%
flat-0%	0.00	100%	0.00	100%

Out-of-sample open-loop replay (audit only, over the 1991→2022 transitions): S1 264.5 and CPM 260.6 × 10^3 acre-ft yr^{-1}; closed-loop from the observed 1990 head, S1 prescribes 268.5 with the cut active 24.2% of years and ends 2023 at 643.3 ft (observed 635.7; training-mean pumping ends 639.2). Every flat policy prescribes its cap throughout, so its supplies coincide by construction. The closed loop is a model object: run from the observed 1934 head it ends 1990 at 677.6 ft against the observed 645.8 ft (+31.8 ft), so the closed-loop supplies carry the map's own level bias and are reported as the audit requested — as the within-model check on the hybrid margin, not as a second retention leg.

Table 4. The retention margins under every comparator reading (open-loop supplies; the closed-loop margins follow in the text).

Comparator (matched protection at 618 ft / UC-min)	Supply	S1 margin	CPM margin
kernel-matched flat-90% (declared grid)	253.94	+3.3%	+0.4%
attractor twin flat-80% (S1) / flat-60% (CPM)	225.73 / 169.29	+16.2%	+50.6%
interpolated 7.2% cut (outside family)	261.84	+0.2%	−2.6% (fails)
flat-92% (1% grid)	259.59	+1.1%	−1.8% (fails)
flat-93% (1% grid)	262.41	−0.02% (fails)	−2.9% (fails)

- **S1: retained (nominal, UC-min, 618 ft; hybrid).** It matches the flat caps' robust invariance (kernel = whole safe set at all horizons) while supplying 262.36 versus flat-90%'s 253.94 (+8.4 × 10^3 acre-ft yr^{-1}, +3.3%) and flat-80%'s 225.73 (+36.6, +16.2%). The scoring regimes are the hybrid declared in Section 2.4 — protection under the perpetual worst-year recharge floor, supply as the 1934–1990 historical replay mean. The reading of the retention is exactly "worst-case protection at historical-mean entitlement," not a within-scenario dominance. Under the floor itself S1 is identically flat-80% (the cut is active every year, and its floor-attractor supply equals flat-80%'s), so the supply margin exists only in wet years that the robust class excludes. Against flat-90% the floor comparison is the sharper one — S1 is always −20% where flat-90% is always −10%, at identical invariance. The reactive architecture justifies its additional structure on the hybrid criterion, not on the robust criterion alone, and the comparator table shows how grid-dependent the margin is: against the 1%-grid smallest securing cap it is +1.1%, and against the interpolated 7.2% cut it is +0.2%.
- **CPM: retained (nominal, same threshold and class).** Attractor 628.36 ft (equal to flat-60%'s) at supply 254.93 versus flat-60%'s 169.29 (+50.6%) and flat-90%'s 253.94 (+0.4%). The +50.6% is priced substantially off Stage IV, a stage occupied in 1 of 90 years.
- **Closed-loop re-check (post-freeze layer).** With each policy in force and the head simulated by the map under actual recharge, S1's supply falls to 258.98 (−1.3% versus the open-loop replay; the cut is active 41.1% of years, not 35.1%, because the map's trajectory dips below 660 ft more often than the observed record did) and CPM's to 254.70 (−0.1%). The hybrid margins attenuate but survive: S1 +2.0% versus flat-90% (+14.7% versus flat-80%), CPM +50.4% versus flat-60% (+0.3% versus flat-90%). The margin is a real property of the rule's wet-year behavior on this map, not an artefact of evaluating the rules on the observed (unregulated-era) heads alone.
- **Certified level:** certified retention is horizon-truncated. At the one tabulated certified horizon the reactive rules inherit training-mean pumping's boundary (706.7 ft, Section 3.3) and fail the certified re-check. The supply figures are nominal-level comparisons, not certified retention.
- **Institutional threshold: nothing retained.** The reactive rules are identically training-mean pumping at every horizon (trigger-on-boundary invisibility, Section 3.2). The flat caps are more protective at finite horizons and extend the viable horizon. No policy meets the retention test there.

- **Current-baseline sensitivity (post-freeze, analytic).** The declared family and every supply figure are relative to the 1934–1990 training mean (282.16). On the 1991–2023 mean actual pumpage (382.16 $\times$ 10^3 acre-ft yr^{-1}), the same attractor arithmetic gives training-mean-pumping 604.52 ft, the securing cut ≈31.5%, and the Stage-I/flat-80% attractor 613.08 ft — below 618 ft, so the 20% sketch does not secure the physical threshold at current use. Re-expression of the family on the current baseline is a registered follow-up, not a scored object here.

3.5 Classification and stress replays

The T = 5 nominal kernel (UC-min, 618 ft) classifies the actual record. Training-mean pumping excludes exactly one actual year from its viable set — 1956, the drought-of-record year (623.15 ft annual mean, above 618 ft and above the 615.72-ft attractor: 1956 is excluded as a *starting* head whose 5-year forward path under perpetual 1956 recharge falls through the threshold, not because the 1956 mean was below 618; its daily minimum, 612.51 ft, was). S1 and CPM exclude none: the entire 90-year actual record is robustly 5-year viable under the cut rules. The T = 5 certified kernels are empty, so no actual year is certified 5-year viable under any policy. This is the boundary of the certified analysis.

A 1950s open-loop diagnostic (model driven by actual R, P versus actual heads, 1951–1956) records the map's drought bias. The affine map under-predicts the decline: model 659.5 → 631.3 versus actual 659.5 → 623.2 ft, maximum error 8.1 ft, biased high. The model-based policy replays from the observed 1950 head keep all policies above 618 ft (BAU minimum 629.7, S1 634.9, CPM 637.1 ft) — note that this replay holds pumping at the training mean (282.16) while actual 1956 pumping was ≈321 $\times$ 10^3 acre-ft, so it is not the decade's actual policy either. The open-loop bias means the true margins are smaller than the replay suggests. This is recorded, and no correction is applied.

3.6 Finite-duration recharge floors and floor-class supply (post-freeze layers)

Two declared post-freeze layers extend the persistent-floor object. The first asks how long a drought-class episode must last before the safe set contracts. The class floor holds for n years, recharge then returns to its training mean, and the infinite-horizon lower boundary at the 618-ft threshold is pulled back by exact backward recursion through the n floor years. The second computes the floor-class supply — the closed loop from the observed 1934 head with recharge held at each class floor — supplying the declared-scoring half that the historical replay cannot.

Table 5. Finite-duration floors (post-freeze): infinite-horizon lower boundary (ft) at the 618-ft threshold after n years at the class floor followed by training-mean recharge. The BAU row at $n = 5$ and $n = 10$ reproduces the registered $T = 5$ and $T = 10$ nominal boundaries (625.6 and 658.4 ft). The $n = 15$ emptiness reproduces the registered 13-year horizon bound.

Policy	UC_min, n=5	UC_min, n=10	UC_min, n=15	UC_q05/q10, any n
BAU	625.6	658.4	empty	618.0
flat-90% and deeper; S1; CPM	618.0	618.0	618.0	618.0

Duration differentiates exactly one policy. Under the drought-class floor, training-mean pumping is the only policy whose boundary moves with episode length. Every cut policy holds the whole safe set at every duration, because the cuts' floor attractors sit at or above the threshold (618.9–647.3 ft, Section 3.1, Table 1). A finite drought episode followed by normal recharge pulls nothing back beyond 618 ft. Under the two milder classes the safe set is already invariant at BAU (Section 3.2), and duration is inert for every policy.

Table 6. Floor-class supply (post-freeze): closed loop from the observed 1934 head (670.4 ft), recharge held at the class floor, mean prescribed pumping (10^3 acre-ft yr^{-1}) over the training span (1934–1990) and the full span (1934–2023); end head at 2023.

Policy	UC_min supply (train/full)	UC_min end head	UC_q05 supply (train)	UC_q10 supply (train)
BAU	282.16 / 282.16	615.7	282.16	282.16
flat-90%	253.94 / 253.94	618.9	253.94	253.94
flat-80%	225.73 / 225.73	622.0	225.73	225.73
S1	226.72 / 226.36	622.0	226.72	226.72
flat-60%	169.30 / 169.30	628.4	169.30	169.30
CPM	174.49 / 172.59	628.4	187.36	187.61
flat-0	0 / 0	647.3	0	0

The table quantifies the scoring-regime reading of Section 2.4. Under the floors the reactive rules converge to their matched flat-cap attractors (S1 to flat-80%'s 622.04 ft, CPM to flat-60%'s 628.36 ft — the cut active every year on the attractor branch). Their closed-loop span-mean supply exceeds the matched caps only by the trigger-lag margin of the descent from the observed 1934 head — S1 +0.4% over flat-80% (226.7 versus 225.7 × 10^3 acre-ft yr^{-1}, a floor-class trigger lag), CPM +3.1% over flat-60% (174.5 versus 169.3). These are different objects from the hybrid margins of Table 4: S1's +0.4% here is the within-floor trigger lag, CPM's +0.4% there is the historical-replay margin against flat-90% — same number, different quantities, and neither is a robust-supply statement. The hybrid criterion's margins exist only in the wet years that the robust classes exclude; the floor-class supply leg is the margin's own measure of how little it would buy inside the class.

3.7 Attractor uncertainty (post-freeze layer)

Table 7. Residual bootstrap of the 1934–1990 fit (5,000 replications; seed 20260905; all refits contractive): policy attractors under UC-min. The last column gives conditional bootstrap shares at or above 618 ft and 660 ft, respectively.

Policy	Point attractor (ft)	Bootstrap mean (ft)	Bootstrap SD (ft)	5th–95th percentile (ft)	Refit share: $H^* \geq 618$ / $H^* \geq 660$
BAU (training-mean)	615.72	613.72	13.09	590.0 – 630.6	0.4148 / 0.0000
flat-90%	618.88	617.00	12.46	594.5 – 633.0	0.5336 / 0.0000
flat-80%	622.04	620.27	11.87	598.7 – 635.7	0.6500 / 0.0000
S1	622.04	620.27	11.87	598.7 – 635.7	0.6500 / 0.0000
flat-70%	625.20	623.55	11.35	603.1 – 638.3	0.7486 / 0.0000
flat-60%	628.36	626.83	10.89	607.3 – 641.2	0.8290 / 0.0000
CPM	628.36	626.19	10.32	607.3 – 639.7	0.8290 / 0.0000
flat-50%	631.52	630.11	10.51	611.3 – 644.2	0.8864 / 0.0000
flat-0% (zero pumping)	647.32	646.49	10.03	629.0 – 661.2	0.9894 / 0.0688

The audit's resolvability question has a quantitative answer. The *ordering* of the attractors is resolved: the flat-90%-minus-BAU gap has a 90% bootstrap band of [1.9, 5.0] ft, excluding zero. The *threshold clearance* is less resolved: the conditional bootstrap shares at or above 618 ft are 0.4148 for training-mean pumping, 0.5336 for flat-90%, and 0.6500 for S1/flat-80%; only cuts of 60% and deeper exceed a 0.8 share, and zero pumping reaches 0.9894. These are refit shares under the stated iid residual bootstrap, not probabilities of real-world threshold attainment. The nominal-level statement that a 10% cut secures the threshold rests on a 2.3-ft gap (615.72 ft to the 618-ft threshold; the marginal flat-90% attractor sits at 618.88 ft) on a map with 5.6-ft residual SD and an acknowledged 8.1-ft high bias in the drought it is scored on — the fitted map cannot resolve whether the marginal members of the family protect 618 ft. The deep cuts and the failure of training-mean pumping are the resolved parts of the result.

The 660-ft extension changes the interpretation of the universal negative result. At UC-min, the point-fit zero-pumping attractor is 647.32 ft (a 12.68-ft positive margin), but its 5th, median, and 95th bootstrap percentiles are 628.99, 647.33, and 661.16 ft; 344 of 5,000 refits (6.88%) place it at or above 660 ft. All refits remain contractive with a negative pumping coefficient. None of the evaluated positive-pumping policies has an iterated attractor at or above 660 ft in these refits; zero pumping is the sole tested rule with a nonzero share. Thus the algebraic comparison theorem remains exact conditional on its coefficients, but the positive-margin hypothesis at 660 ft is not robust to this coefficient resampling. These are model-conditional shares, not probabilities of real-world safety; this post-freeze check does not recompute viability kernels or alter the frozen retention score.

4. Discussion

The complete evaluation loop — measured state, calibrated stock-flow map, declared governance operators, declared uncertainty classes, viability kernels with explicit declared-defect erosion, a held-out defect audit, and a fixed retention rule — yields a **nominal selection at the physical threshold under the mildest floor class**. The reactive architecture matches flat-cap protection at 3.3% (S1) and 0.4% (cascade) more permitted supply than the kernel-matched flat-90% cap — 16.2% and 50.6% against their attractor twins, +1.1% and −1.8% against the 1%-grid smallest securing cap — under a hybrid criterion whose wet-year entitlement the closed-loop re-check attenuates but does not remove. An interpolated 7.2% mean cut (outside the declared family; about 31.5% of current use) secures the physical threshold against a perpetual drought-of-record where training-mean pumping fails within roughly 13 years (a ceiling-relative statement, Section 3.2). Two boundaries are equally part of the result. At the point-fit coefficients, the 660-ft safe set is not invariant under the declared classes; the fixed-map margin argument extends this result to every non-negative pumping sequence when its positive-margin condition holds. That empirical premise is not coefficient-robust: under UC-min, the zero-pumping attractor is at or above 660 ft in 6.88% of residual- bootstrap refits (Section 3.7). This post-freeze sensitivity does not re-score the frozen retention rule. Separately, the certified content is defect-bound to $T \leq 3$ years.

The reactive result is system-dependent, not architectural. Its mechanism is the aquifer's physics. High transmissivity and rapid karst recharge make wet-year recovery fast, so a flat cap permanently taxes recoveries that a reactive rule harvests — the supply margin exists only in the wet years that the robust classes exclude (Section 3.4). A companion intervention study on Northern cod applies the identical design and retains nothing: there the reactive rules cut catch exactly where the moratorium already protects. The framework's deliverable is the scored comparison itself; which governance architecture earns its complexity is a property of the system, and the two scored systems occupy opposite ends of that design space without bounding it.

Two layers of results must be kept distinct. The robust-kernel findings are statements about the declared pumping family under the declared floor classes. The certified-kernel findings are a

different layer, produced by applying the erosion conversion to the declared training defect. With $\varepsilon = 15.41$ ft — exceeded out of sample at 21.81 ft — every positive-pumping policy's certified kernel is empty beyond T = 3 years, and the bound is optimistic rather than conservative on this object (Section 3.3). The certified emptiness is a bound on what the fitted map can certify, not a robustness statement about the physical system. The mechanism of the reactive rules' certified collapse is geometric — the eroded threshold enters the trigger band (Section 2.4) — and the certified comparison is conservative in form against state-dependent rules for that reason.

The 660-ft design observation also reads correctly against the institution it describes. The actual CPM rule manages **frequency** — the fraction of time the head spends in critical stages — not robust invariance. The point-fit map's no-invariance result is consistent with that design intent: the rule cannot make wet years, it prices dry ones. The post-freeze bootstrap qualifies the parameter certainty of that result: the zero-pumping attractor is at or above 660 ft in 6.88% of UC-min refits, while no positive-pumping policy in the evaluated family does so in these draws. This is a fixed-map sensitivity, not a performance evaluation of stage frequency or springflow. The National Academies' review of the Habitat Conservation Plan (National Research Council 2015) was likewise organized around monitoring and modeling adequacy under drought. The horizon-not-invariance distinction remains the robust-kernel expression of that institutional boundary.

The karst honesty of the object is inherited from its forecast-study companion and from the modeling lineage it descends from. The one-pool affine map is the simplest member of the lumped family whose regional adequacy was tested on the Barton Springs segment (Scanlon et al. 2003; Hutchison and Hill 2011). Conduits, the Uvalde–San Antonio divide, and unconfined recharge-zone storage remain in the residual. San Antonio and Uvalde recharge and pumpage are lumped. The actual CPM triggers are 10-day averages, so the annual-mean rule is a coarse relative of the real institution. Stages II–IV are declared scenarios, not verified institutions. Nominal kernels carry no defect margin, and the certified kernels use a training defect that the out-of-sample audit exceeds. The counterfactual swap from the largely unregulated fitting era to state-dependent rules carries the declared econometric boundary of Section 2.1 (Lucas 1976). The UC floors are certification geometry, not forecasts. The 1950s replay is biased high by 8.1 ft. Nothing in this leg promotes or demotes any forecast module, and no two-pool, karst, or solute claim is made.

Cross-system reading. The protocol is the deliverable, and its value is shown by contrast rather than by a single result. The identical intervention-selection design applied to Northern cod (Abaee, 2026b) retains nothing — there the reactive rules cut catch exactly where the moratorium already protects, so no policy earns its complexity. Here, on the Edwards Aquifer, the reactive architecture earns a *nominal* supply margin at the physical threshold (+3.3% Stage I, +0.4% cascade) that is real but wet-year-contingent. The two scored systems therefore occupy opposite ends of the design space — one where reactive complexity is unrewarded, one where it is marginally rewarded — and the framework does not prescribe which end a new system lands on; it shows how to find out. That is the general, transferable content: the method scores protection against the robust floor class and separates the geometric (kernel/ threshold) from the identified (defect) findings, independent of the specific system.

Practical reading. For a district that already operates triggers, the result gives two usable bounds. First, at the physical threshold, deep cuts are robust and the smallest securing cut is interpolated at 7.2% of training-mean pumping (about 31.5% of current use) against a perpetual drought-of-record — but a cap's nominal supply advantage over a reactive rule is not robust under the declared floor classes; replayed wet-year supply is a hybrid-score contribution, not a robustly available margin. Second, at the institutional (660-ft) threshold, the point-fit map under the persistent drought-of-record floor admits no invariant safe set, even at zero pumping. The post-freeze UC-min bootstrap places the zero-pumping attractor at or above 660 ft in 6.88%

of refits; thus the positive-margin premise is not coefficient-robust. No positive-pumping policy in the evaluated family has an iterated attractor at or above that threshold in these refits, and the analysis does not exhaust all possible state-dependent sequences. The sensitivity is conditional on the fitted map and disturbance class, not an estimate of how the actual 10-day CPM rule changes stage frequency or springflow. Where certified claims are wanted (a formal finite-horizon guarantee), the statement is limited to $T \leq 3$ years by the fitted defect (train $\varepsilon = 15.4$ ft, exceeded out of sample at 21.8 ft), so any longer certified claim should be read as a bound, not a certificate.

Generalizability boundary. The protocol generalizes; the verdicts do not. Every numerical certificate is scoped to the J-17 annual-mean one-pool affine map and the declared governance operators and floor classes, and no transfer to the Uvalde Pool (J-27) is claimed, consistent with the programme's no-pooling discipline. Conduits, the Uvalde–San Antonio divide, and unconfined recharge-zone storage remain in the residual, and the annual-mean trigger is a coarse relative of the real 10-day rule.

Two senses of generality should be kept apart here, because the margin argument of Section 3.2 is general in one of them and not the other. Within *this* calibrated map, the theorem is general over *rules*, conditional on its hypotheses: when $0 < a < 1$, $\gamma < 0$, and the zero-pumping attractor lies below the threshold, no non-negative pumping sequence can keep the threshold set invariant. That proof is exact for each coefficient vector satisfying those conditions, so richer triggers or hysteresis cannot evade it when the positive-margin premise holds. But the UC-min bootstrap shows that this premise is not coefficient-robust at 660 ft: 6.88% of refits put the zero-pumping attractor at or above the threshold. The numerical verdict therefore remains conditional on the fitted vector and is not a parameter-robust statement. It says nothing about *other aquifers*: whether any particular pool has a positive margin at its own threshold depends on that pool's fitted $(a, \alpha, \beta, \gamma)$ and its own floor, which must be estimated there and reported with the hypotheses checked rather than assumed. Generality across rules is conditional; generality across pools is not claimed.

5. Conclusions

On the measured J-17 record, three verdicts, not one. (1) **Nominal, physical threshold, mildest floor class:** reactive trigger rules match flat-cap robust protection of the 618-ft threshold at 3.3% (Stage-I sketch) and 0.4% (cascade) more permitted supply than the kernel-matched flat-90% cap — 16.2% and 50.6% against their attractor twins — under a hybrid criterion (worst-case protection, historical-replay entitlement) whose wet-year margin survives a closed-loop re-check attenuated (S1 +2.0% versus flat-90%). An interpolated 7.2% mean cut of training-mean pumping (about 31.5% of current use) secures the threshold against a perpetual drought-of-record, where training-mean pumping fails within roughly 13 years (ceiling-relative). (2) **Certified:** nothing — every positive-pumping certified kernel is empty beyond three years, an optimistic bound on a defect the out-of-sample audit exceeds. (3) **Institutional threshold:** under the persistent drought-of-record floor, the point-fit map admits no invariant 660-ft safe set, even at zero pumping. In the post-freeze UC-min bootstrap, 344/5,000 refits (6.88%) place the zero-pumping attractor at or above 660 ft; the positive-margin condition behind the all-sequences theorem is therefore not coefficient-robust. None of the evaluated positive-pumping policies has an iterated attractor at or above 660 ft in these refits, and no claim is made about every possible time-varying rule. This geometric sensitivity is conditional on the fitted one-pool map and does not estimate the live rule's effect on observed stage frequency or springflow. The fitted map resolves the deep-cut protections and the failure of training-mean pumping, but not the marginal caps (bootstrap shares 0.4148–0.6500); that is the honest resolution limit of the deliverable — the scored comparison on one measured system.

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Declarations

Data Availability Statement

All input data, analysis scripts, and result files are archived in the public repository at https://github.com/MIKEAA2020/general-sustainability. J-17 daily highs: Texas Water Development Board well 6837203. Recharge: Umphres and Choi (2025), USGS data release, https://doi.org/10.5066/P1BI62NY. Pumpage: Edwards Aquifer Authority (2024/25), Table 1. The full intervention protocol — object, defect declaration, disturbance classes, policy family, and retention rule — was frozen and dated 2026-08-26 before any kernel, boundary, replay, or retention score was computed. It is archived with the analysis code as the frozen-protocol record, alongside the companion forecast-evaluation protocols dated 2026-08-25. The analysis is fully deterministic: re-executing the registered runner regenerates both output files, and a verification re-execution in a fresh environment reproduced both byte for byte. The machine-readable outputs include the nominal and certified retention fields and the certified-horizon record. The Section 3.6 layers (finite-duration floors; floor-class supply) are produced by `rerun_campaigns/campaign_e4_elevation.py`, archived alongside their outputs, and regenerate them exactly. The post-freeze audit layers of Sections 3.4 (closed-loop supply replay), 3.4's current-baseline sensitivity, and 3.7 (attractor bootstrap) are produced by `wave_e_edwards/src/e4_audit_layer.py` (seeded, deterministic, on the registered panel and the paper's declared coefficients), with outputs archived as `wave_e_edwards/results/e4_audit_layer.json`. The additional 660-ft coefficient-sensitivity extension is not part of the frozen retention calculation. Its copied annual panel, deterministic runner, and CSV outputs are archived under `paper10b_edwards_aquifer_v3_src/`; the refit shares are conditional on that script's iid residual bootstrap and fixed UC-min floor.

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Competing interests

None.

Code availability

The kernel computation and the policy scoring are produced by the scripts named above, archived with their outputs. The verification script `paperE4_edwards_intervention_v16_verification.py` re-runs the intervention computation from the registered panel and recompares every attractor of Table 1, the smallest securing cut, the certified horizons, and the fit

diagnostics against the recomputed values; it is archived alongside this manuscript. The post-freeze 660-ft sensitivity is reproduced by `paper10b_edwards_aquifer_v3_src/verify_660_attractor_bootstrap.py`; it regenerates the policy and zero-pumping CSVs from the included panel using seed 20260905.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.